

Abdullah Erzin

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Summary

Computer Vision Engineer with ~5 years in robotics and machine vision: camera calibration, stereo depth, 3D coordinate transforms, and real-time detection on edge hardware (NVIDIA Jetson, Intel RealSense). Builds and owns production CV pipelines in Python and C++, from algorithm to field deployment.

Work Experience

Computer Vision Engineer — Pek Automotive

Logatec, Slovenia | Apr 2025 – Present

- Owned the end-to-end perception-to-picking pipeline for autonomous multi-arm apple and orange harvesting robots (camera calibration, object detection, depth estimation, picking logic); brought the previously non-operational system to field-ready operation and served as the responsible CV engineer during field deployments in five countries.
- Designed a three-stage camera-to-arm calibration pipeline (per-arm 2D polynomial regression, linear misalignment correction, camera tilt compensation during arm extension) that replaced a manual procedure, achieving 5 mm max positional error (2 mm std) against a 10 mm production threshold; added outlier rejection, R^2 validation, automatic retry, and UART status reporting so operators can verify results without a PC.
- Improved Intel RealSense D435 depth accuracy by replacing peak-based depth with a histogram front-surface search and fruit pixel masking, reducing systematic overestimation from fruit geometry and foliage; also corrected perspective-projection bias in fruit diameter estimation.
- Implemented saturation-based camera exposure control that adapts faster than default auto-exposure under changing sunlight, reducing overexposed infrared (IR) and depth frames.
- Built a semi-automated annotation workflow from 700 GB of field recordings and retrained YOLO instance segmentation models on occluded fruit clusters, improving detection of partially visible fruit.
- Designed picking logic for clustered fruit, including dual-arm coordination, frontmost-first picking with rescans, and re-identification by diameter and position to prevent duplicate picks and damage to adjacent fruit.
- Owned production deployment for the Intel NUC fleet: automated Ubuntu image builds (autoinstall, post-install provisioning), systemd service management, remote deployment and log collection, and Linux camera driver debugging; built robot simulation mode and field-log replay tools for hardware-free development.

Software Engineer — Robsys Robotic Systems

Istanbul, Turkey | Mar 2024 – Mar 2025

- Built an autonomous navigation system from scratch for a cleaning robot (custom SLAM, pathfinding, obstacle avoidance), since off-the-shelf SLAM frameworks did not run on the target hardware.
- Developed a multi-sensor fusion pipeline on NVIDIA Jetson combining stereo cameras, gyroscopes, and ultrasonic sensors; trained custom deep learning models for environmental perception and wrote low-level firmware for sensor synchronization and motor control.
- Co-designed a UART protocol with an embedded engineer for bidirectional data exchange between the robot's motherboard and its subsystems.
- Integrated real-time telemetry and camera streaming into a mobile app for remote monitoring, control, and safety operations.

AI Developer — Bomensoft

London, UK (Remote) | Jan 2024 – Feb 2024

- Optimized AI models and extended backend and web features (Python, JavaScript).

Computer Vision Engineer — Doğru Holding (DGR Project)

Istanbul, Turkey | May 2023 – Dec 2023

- Developed a face recognition system on Jetson Nano with 98% accuracy, 30% fewer false positives, and face embeddings stored in 4 KB on-device.
- Built a stereo-camera liveness detection module from scratch: extracted per-keypoint facial depth profiles, collected a labeled real vs. spoofed dataset, and trained a lightweight TensorFlow model to reject spoofing attempts.
- Implemented performance-critical components in C++ and exposed them to the Python pipeline as native extensions to run in real time on constrained hardware.
- Built a real-time AI video pipeline that anonymized people in security camera feeds with YOLO-based detection and per-detection overlays, then streamed to YouTube Live at a stable 30 fps; kept per-frame processing under 30 ms by modifying the streaming library to manipulate outgoing frame packets directly.

Autonomous Vehicle Development Engineer — Doğru Holding (ZGN Autonomous & Robotics)

Istanbul, Turkey | Dec 2022 – May 2023

- Optimized the pathfinding algorithm for autonomous vehicles, reducing processing time by 30%.
- Designed an algorithm that lets robots build maps directly from construction blueprints.

Machine Vision Engineer — Gensys Automation & Machine Vision

Kocaeli, Turkey | Dec 2021 – Dec 2022

- Developed machine vision applications in Halcon and C#, integrated into MSSQL-backed desktop tools, and commissioned systems on-site.
- Led a TUBITAK-funded thermal NDT research project: modeled heat diffusion analytically, built a halogen-lamp and thermal-camera test setup, and developed a desktop application that estimated the depth of subsurface defects in PLA parts.
- Developed an automated image-cropping tool to preprocess data for ML training.

Machine Vision Intern — Mavis Machine Vision

Kocaeli, Turkey | Jul 2020 – Mar 2021

- Built image-processing applications in Halcon and C# with SQL database integration.

Education

B.Sc. in Mechatronics Engineering — Marmara University

Istanbul, Turkey | Graduated Mar 2021

Projects

Web-Based AR Face Filter Platform — Side Project

- Real-time multi-face detection and tracking (up to 8 faces) powering 30+ WebGL/Three.js AR filter effects.
- Subscription and device-licensing backend with a Node.js REST API and Supabase (PostgreSQL).
- GitOps pipeline with GitHub Actions, multi-arch Docker builds (ARM64/AMD64), and Watchtower auto-updates on Raspberry Pi devices; multi-stage builds cut image size by 60%.

Skills

- **Languages:** Python, C++, C#, JavaScript, SQL
- **Computer Vision:** Camera calibration, stereo vision, depth sensing, 3D coordinate transforms, object detection and segmentation, OpenCV, YOLO, Halcon
- **Machine Learning:** PyTorch, TensorFlow, NumPy
- **Hardware and Embedded:** NVIDIA Jetson, Intel RealSense, UART, Linux
- **Infrastructure:** Docker, GitHub Actions, Git, Prometheus, Grafana
- **Interests:** Motorcycles, scuba diving (1-star diver)

References available on request.